

LIEW PHING SHUIN

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SUMMARY

AI Engineer specializing in scalable, business-driven AI solutions. Experienced in LLM integration, RAG systems, utilizing AI coding agents, and workflow automation to accelerate software delivery and business operations. Skilled in building end-to-end AI pipelines, including embeddings, vector databases (FAISS, Qdrant), cloud and self-hosted LLM deployments, and agentic workflows. Proficient in Python, FastAPI, and Azure OpenAI, delivering production-ready AI platforms and applications.

EXPERIENCE

AI Engineer

V Gallant (VCIG)

December 2025 - Present, Kuala Lumpur

- Led enterprise automation initiatives and workflow orchestration using n8n, streamlining operational processes and AI-driven workflows.
- Architected, deployed, and operated a self-hosted AI creative generation platform (**Pixel Factory**) serving FLUX 2 image generation and LTX 2.3 video generation models through **vLLM-Omni** inference infrastructure.
- Deployed and managed **DeepSeek V4 Flash** inference infrastructure across dual-GPU servers using **vLLM**, integrating with Open WebUI to deliver scalable Text-as-a-Service (TaaS) capabilities with optimized throughput and low-latency inference.
- Developed an AI-powered invoice processing pipeline using Vision Language Models (VLMs) to extract structured data from unstructured invoices, combined with Playwright browser automation for autonomous POS system submission.
- Built a multi-tenant vector search platform using Qdrant with hybrid retrieval architecture (BGE sparse embeddings, dense embeddings, and reranking) to support high-accuracy Retrieval-Augmented Generation (RAG) systems.
- Engineered a Text-to-SQL service and query routing between structured SQL databases and unstructured RAG pipelines to optimize response accuracy and relevance.
- Designed and implemented an asynchronous embedding and ingestion pipeline with job tracking, background workers, and scalable processing architecture for high-volume AI data workflows.
- Implemented backend services and API integrations to support AI inference, document processing, retrieval systems, workflow automation, and enterprise application connectivity.

AI/ML Engineer

Keysight Technologies

March 2024 - November 2025, Penang

- Implemented video transcription model for chatbot data (ASR) - OpenAI Whisper.
- Set up LLM models locally and serve using llama.cpp with CUDA (GPU optimization).
- **Built a Document AI Chatbot** that processes emails, meeting recordings, and slides, enabling R&D team members and engineers to efficiently retrieve technical insights and project knowledge. (LLM + RAG).
- Built a scalable news scraping tool to extract article content (**Selenium** and **BeautifulSoup**).
- Automated news processing (titles, keywords, summaries) using **LLM**, reducing manual workload by 70% for analysts.
- Automated insights extraction and report generation from PDF reports using **LLM**.
- Developed product recommender system (**Association Rule Mining** + **Collaborative Filtering**).
- Built interactive web applications with **Streamlit** to test and demonstrate AI/ML prototypes.

Market Intelligence Analyst

Keysight Technologies

August 2023 - January 2024, Penang, Malaysia

- Analytics, prediction and forecasting projects using statistical method and various Machine Learning algorithms.
- Developed time series forecasting models to predict economic indicators and market trends.
- Built data models for an early alert system to anticipate business and market downturns and recoveries.
- Researched an inflation simulator model for forecasting inflation scenarios.
- Explored Market Basket Analysis techniques to uncover product association rules and enhance cross-selling strategies.
- Volunteered with KEYS2School and GETS (Girls in Engineering & Technologies) to support STEM education initiatives.

PROJECT

Quantus (Trading alert bot)

- A Python market-scanning bot that detects double-bottom "flush-reclaim" chart patterns and pushes annotated Telegram/Discord alerts — with a walk-forward backtesting engine, SQLite state, cron/launchd scheduling, and automated tests.
- Designed and backtested a systematic bear-trap reversal equity strategy in Python: walk-forward simulation with in-sample/out-of-sample validation, parameter sweeps, and R-multiple / profit-factor / drawdown analytics with ATR-based signal filters and risk-based position sizing.

EDUCATION

Bachelor of Management (Honours), Major in Business Analytics
Minor in Computer Science · University of Science, Malaysia · Penang, Malaysia · 2024 · CGPA 3.70 (First Class Honours)

Pre-University Programme
Matriculation College Negeri Sembilan · Negeri Sembilan, Malaysia · 2020 · 4.0

Secondary Education (Science Stream)
SMK (CF) Keat Hwa · Kedah, Malaysia · 2018 · 10A's

CERTIFICATIONS

Claude Code in Action
Anthropic · 2026

Fundamentals of AI Agents Using RAG and LangChain
IBM · 2025

Docker Fundamentals
packt · 2025

Generative AI for Data Scientist Specialization
IBM (Coursera) · 2024

IBM Python for Data Science, AI & Development
2024

Introduction to TensorFlow for AI, ML, and Deep Learning
Deep Learning.AI · 2024

SKILLS

AI / LLM Engineering: RAG · Agentic workflows · LLM serving & inference (vLLM, vLLM-Omni, llama.cpp, Open WebUI) · Vision Language Models (VLMs) · Text-to-SQL · ASR (Whisper) · LangChain · Azure OpenAI · Prompt engineering

Languages: Python · SQL · C++ · C# · R

Backend & Data Engineering: FastAPI · REST API design & integration · Async pipelines & background workers · Playwright · Selenium · BeautifulSoup · Streamlit

ML & Data Science: PyTorch · TensorFlow · scikit-learn · Time-series forecasting · NLP & sentiment analysis

Infrastructure & Tooling: Docker · CUDA / GPU optimization · Vector databases (Qdrant, FAISS) · n8n · Git

Data Visualization: Power BI · Tableau